IT-SIMPLERA INSTITUTE

Network Administration Internship Program

Week 03 Report — Enterprise Multi-Protocol Routing Challenge

(OSPF Multi-Area, Static Routing & RIP v1/v2) — Built and Verified in GNS3

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Week	03
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Tools Used	Cisco Packet Tracer 8.2.1, GNS3 2.2.40.1, PuTTY

1. Table of Contents

2. 1. Objectives

This project designs, implements, and troubleshoots a medium-sized enterprise network in GNS3 (Cisco 3725 routers) combining OSPF Multi-Area routing, RIP v1/v2, and static routing. The goals are to:

- Design a scalable, VLSM-addressed enterprise topology across six routers, four LAN switches, and eight end devices (VPCS).
- Configure and verify OSPF Multi-Area routing (Area 0 backbone + Area 1).
- Configure and verify RIP v1, then migrate to RIP v2, observing the classful vs. classless behavior difference.
- Configure static routing for a branch/stub router (R5).
- Implement mutual route redistribution between OSPF, RIP, and Static domains.
- Verify end-to-end reachability with ping/traceroute across all four LAN segments, and document real troubleshooting encountered during the build.

3. 2. Network Design Overview

The topology uses six Cisco 3725 routers (R3, R4, R5, R6 fitted with an additional NM-1FE-TX module to provide a third FastEthernet interface) connected in a chain, with three routing domains joined through redistribution:

- OSPF domain: R1 (Area 1) — R2 (ABR) — R3 (Area 0), router IDs assigned via loopback interfaces.
- RIP v2 domain: R3 — R4 — R6, initially configured as RIP v1 then upgraded to RIP v2 to correctly support the VLSM-subnetted links.
- Static domain: R5 is a stub/branch router reachable only via a static route configured on R6, with a default route back to R6 configured on R5.

R3 acts as the ASBR, redistributing RIP routes into OSPF and OSPF routes into RIP. R6 additionally redistributes the static route to R5's LAN into RIP, so that R5's network becomes reachable across the entire topology (RIP domain and, via R3, the OSPF domain as well).

Figure 1: Final working topology as built and verified in GNS3.

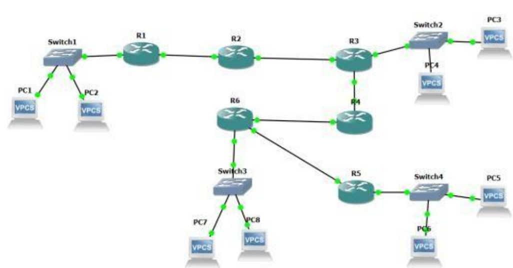


Figure 1b: Actual GNS3 canvas screenshot — six-router topology with Switch1–4 and PC1–PC8, as built and verified.

4. 3. IP Addressing Table (VLSM)

Base network block: 192.168.10.0/24, subnetted with VLSM.

3.1 LAN Subnets

Segment	Network	Mask	Usable Range	Gateway
LAN1 (R1) - Switch1	192.168.10.0/28	255.255.255.240	.1 - .14	192.168.10.1
LAN3 (R3) - Switch2	192.168.10.16/28	255.255.255.240	.17 - .30	192.168.10.17
LAN5 (R5) - Switch4	192.168.10.32/28	255.255.255.240	.33 - .46	192.168.10.33
LAN6 (R6) - Switch3	192.168.10.48/28	255.255.255.240	.49 - .62	192.168.10.49

3.2 Point-to-Point WAN Links (/30)

Link	Network	Router A	Router B	Purpose
R1-R2	192.168.10.64/30	.65 (R1 Fa0/1)	.66 (R2 Fa0/0)	OSPF Area 1
R2-R3	192.168.10.68/30	.69 (R2 Fa0/1)	.70 (R3 Fa0/0)	OSPF Area 0
R3-R4	192.168.10.72/30	.73 (R3 Fa0/1)	.74 (R4 Fa0/0)	RIP v1 -> v2 domain
R4-R6	192.168.10.84/30	.85 (R4 Fa1/0)	.86 (R6 Fa0/1)	RIP v2 domain
R6-R5	192.168.10.80/30	.82 (R6 Fa0/0)	.81 (R5 Fa0/0)	Static routing boundary

3.3 Loopback Interfaces (Router IDs)

Router	Loopback0 Address
R1	1.1.1.1/32
R2	2.2.2.2/32
R3	3.3.3.3/32
R4	4.4.4.4/32
R5	5.5.5.5/32
R6	6.6.6.6/32

5. 4. Device Role Summary

Router	Role	Protocols Running
R1	OSPF Area 1 internal router	OSPF
R2	OSPF ABR (Area1 <-> Area0), transit only, no LAN	OSPF
R3	OSPF Area 0 router + ASBR (RIP<->OSPF redistribution)	OSPF, RIP

R4	RIP v2 transit router, no LAN (Fa0/1 unused/shutdown)	RIP
R5	Static-only branch/stub router with LAN5	Static
R6	RIP v2 edge router + static boundary to R5 + LAN6	RIP, Static

6. 5. Router Configurations

Final working configurations for all six routers, as applied and verified in GNS3 (Cisco 3725 with IOS console access via SolarWinds Solar-PuTTY).

5.1 R1 — OSPF Area 1

```
hostname R1 ! interface Loopback0
ip address 1.1.1.1 255.255.255.255
!
interface FastEthernet0/0
 description LAN1 -> Switch1 (PC1, PC2)
ip address 192.168.10.1 255.255.255.240
no shutdown ! interface FastEthernet0/1
description Link to R2
 ip address 192.168.10.65
255.255.255.252  no shutdown !
router ospf 1  router-
id 1.1.1.1
 network 192.168.10.0 0.0.0.15 area
1  network 192.168.10.64 0.0.0.3 area
1  network 1.1.1.1 0.0.0.0 area 1 end
```

5.2 R2 — OSPF ABR

```
hostname R2 ! interface Loopback0  ip
address 2.2.2.2 255.255.255.255 !
interface FastEthernet0/0  description
Link to R1 (Area 1)  ip address
192.168.10.66 255.255.255.252  no
shutdown ! interface FastEthernet0/1
description Link to R3 (Area 0)  ip
address 192.168.10.69 255.255.255.252  no
shutdown !
router ospf 1  router-
id 2.2.2.2
 network 192.168.10.64 0.0.0.3 area
1  network 192.168.10.68 0.0.0.3 area
0  network 2.2.2.2 0.0.0.0 area 0 end
```

5.3 R3 — OSPF Area 0 / ASBR

```
hostname R3
 !
```

```
interface Loopback0
 ip address 3.3.3.3 255.255.255.255 !
interface FastEthernet0/0  description
Link to R2 (Area 0)  ip address
192.168.10.70 255.255.255.252
 no shutdown ! interface
FastEthernet0/1
 description Link to R4 (RIP domain
boundary)  ip address 192.168.10.73
255.255.255.252  no shutdown ! interface
FastEthernet1/0
 description LAN3 -> Switch2 (PC3, PC4)
ip address 192.168.10.17 255.255.255.240
no shutdown ! router ospf 1  router-id
3.3.3.3
 network 192.168.10.68 0.0.0.3 area 0  network
192.168.10.16 0.0.0.15 area 0  network 3.3.3.3
0.0.0.0 area 0  redistribute rip metric 20
metric-type 1 subnets ! router rip  version 2
network 192.168.10.72  redistribute ospf 1 metric
3  no auto-summary end
```

5.4 R4 — RIP v2 Transit

```
hostname R4 !
interface Loopback0  ip address
4.4.4.4 255.255.255.255 ! interface
FastEthernet0/0  description Link
to R3
 ip address 192.168.10.74
255.255.255.252  no shutdown ! interface
FastEthernet0/1  shutdown ! interface
FastEthernet1/0
 description Link to R6 (RIP domain)  ip
address 192.168.10.85 255.255.255.252  no
shutdown !
```

```
router rip
version 2 network
192.168.10.72
network
192.168.10.84
network 4.4.4.4
no auto-summary
end
```

5.5 R6 — RIP v2 Edge + Static Boundary

```
hostname R6 !
interface Loopback0  ip address
6.6.6.6 255.255.255.255 ! interface
FastEthernet0/0
 description Link to R5 (static
boundary)  ip address 192.168.10.82
255.255.255.252  no shutdown ! interface
FastEthernet0/1
 description Link to R4 (RIP domain)  ip
address 192.168.10.86 255.255.255.252  no
shutdown ! interface FastEthernet1/0
 description LAN6 -> Switch3 (PC7, PC8)
ip address 192.168.10.49 255.255.255.240
no shutdown !
router rip
version 2
 network 192.168.10.84
network 192.168.10.48
network 6.6.6.6
 redistribute static metric
2  no auto-summary !
ip route 192.168.10.32 255.255.255.240 192.168.10.81 end
```

5.6 R5 — Static-only Branch Router

```
hostname R5 ! interface Loopback0  ip
address 5.5.5.5 255.255.255.255 !
interface FastEthernet0/0  description
Primary link to R6  ip address
192.168.10.81 255.255.255.252  no
shutdown !
interface FastEthernet1/0  description
LAN5 -> Switch4 (PC5, PC6)
```

```
 ip address 192.168.10.33 255.255.255.240
 no shutdown !
ip route 0.0.0.0 0.0.0.0 192.168.10.82 end
```

---

7. 6. RIP v1 to v2 Migration

RIP v1 is classful and does not carry subnet mask information in its updates, so it cannot correctly represent the VLSM-subnetted links used in this design. R3 and R4 were first configured with plain "router rip" (defaulting to v1) and verified with "show ip route rip" — the VLSM subnets were not learned correctly. Both routers were then upgraded with "version 2" and "no auto-summary", after which "show ip route rip" showed all subnets correctly, confirming RIPv2's classless behavior.

8. 7. Route Redistribution

Two redistribution points exist in this topology:

- R3 (ASBR): "redistribute rip metric 20 metric-type 1 subnets" under OSPF brings RIP-learned routes into OSPF as External Type-1 routes; "redistribute ospf 1 metric 3" under RIP brings OSPF routes into the RIP domain.
- R6: "redistribute static metric 2" under RIP brings the static route to R5's LAN (192.168.10.32/28) into the RIP domain, from where R3 redistributes it further into OSPF — making R5's LAN reachable from R1 and R3 as well.

Verify with "show ip route" — look for R (RIP), O E1 (OSPF external from RIP), and S (static) markers propagating correctly across all six routers.

9. 8. Verification Plan

- show ip ospf neighbor — FULL adjacency confirmed on R1-R2 and R2-R3.
- show ip route rip — verified on R3, R4, R6 before and after the RIPv1->v2 migration.
- show ip protocols — confirmed RIP v2 send/receive on R3, R4, R6.
- show ip route — confirmed on all six routers; R5's LAN and R6's LAN both visible network-wide via redistribution.
- End-to-end ping and traceroute from PC1 (LAN1) to PC3/PC4 (LAN3), PC5/PC6 (LAN5), and PC7/PC8 (LAN6) — all confirmed reachable.

Verification Screenshots

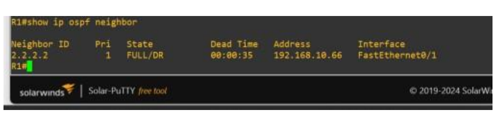
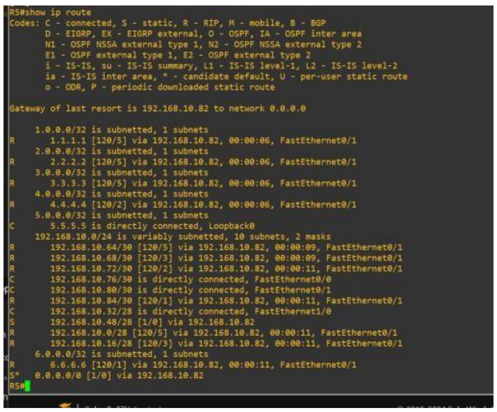


Figure 8.1: R1 – OSPF neighbor table (FULL adjacency with R2).

Figure 8.2: R3 – OSPF neighbor table (FULL adjacency with R2).



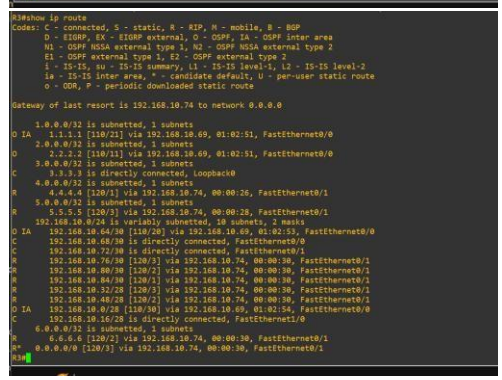


Figure 8.3: R1 – full routing table (OSPF Area 1 + redistributed routes).

Figure 8.4: R3 – full routing table (OSPF + RIP + redistributed routes).

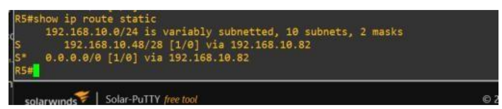


Figure 8.6: R5 – show ip route static (static routes only).

Figure 8.5: R5 – full routing table (default route via R6).

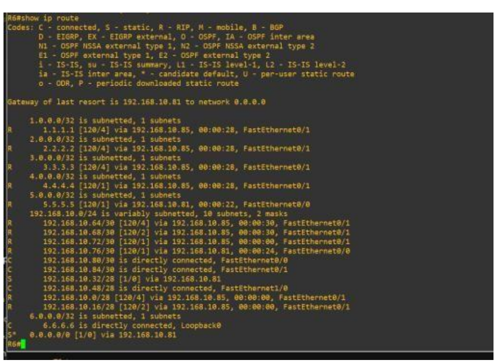


Figure 8.7: R6 – full routing table (RIP + static routes).

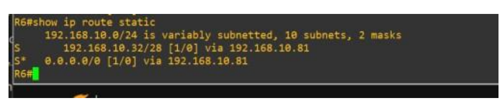


Figure 8.8: R6 – show ip route static (static route to R5's LAN).

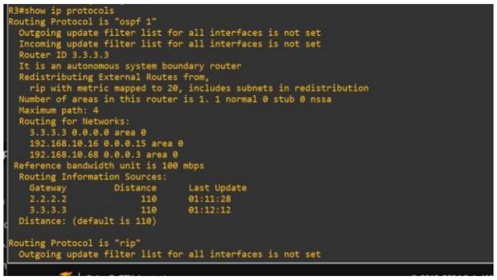


Figure 8.9: R3 – show ip protocols, OSPF process (ASBR + redistribution).

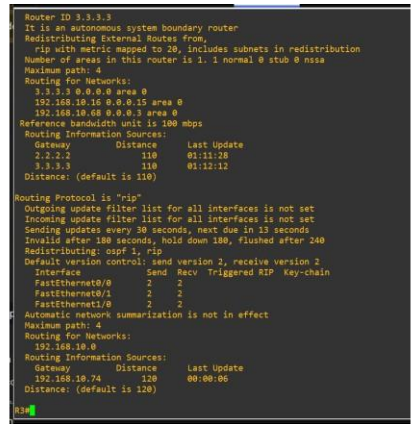


Figure 8.10: R3 – show ip protocols, RIP process (v2 send/receive).

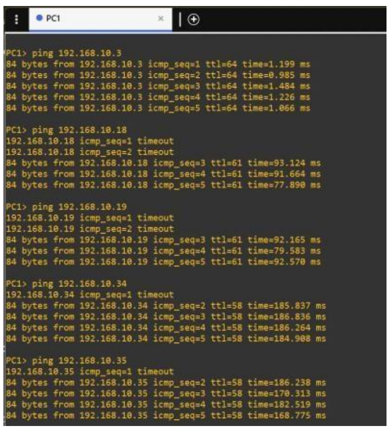


Figure 8.11: PC1 – ping results to LAN3 and LAN5 hosts (192.168.10.18/.19/.34/.35).

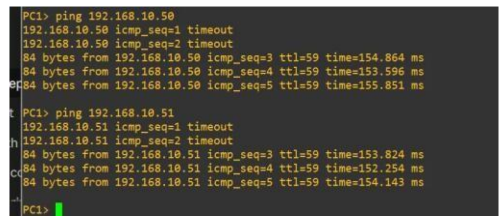


Figure 8.12: PC1 – ping results continued (192.168.10.50/.51, LAN6).

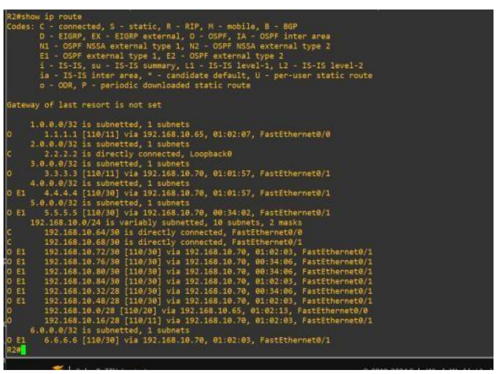


Figure 8.13: R2 – full routing table (OSPF Area 1/Area 0 backbone, no LAN, transit only).

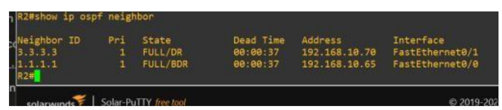


Figure 8.14: R2 – OSPF neighbor table (FULL adjacency with R1 and R3).

Figure 8.15: R3 – show ip route rip (RIP-learned routes on the ASBR).

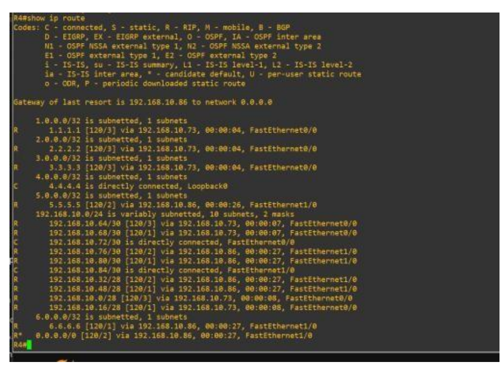


Figure 8.16: R4 – full routing table (RIP v2 transit router, gateway of last resort via R6).

Figure 8.17: R4 – show ip protocols, RIP process (v2 send/receive on all interfaces).

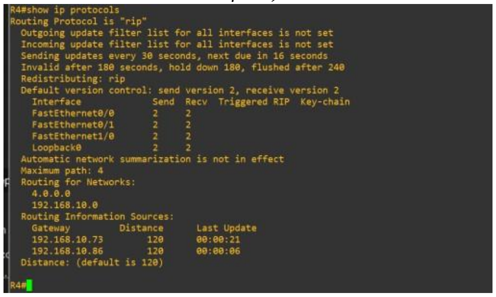
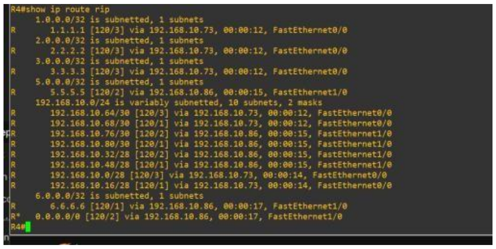


Figure 8.18: R4 – show ip route rip (RIP routes learned from R3 and R6).



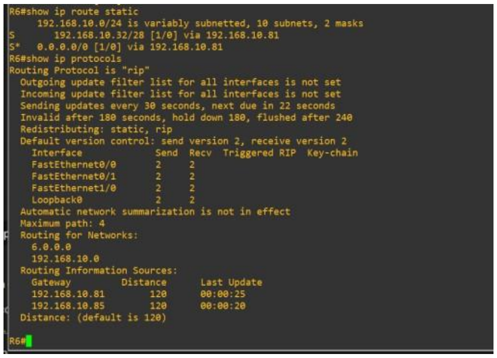


Figure 8.19: R6 – show ip route static and show ip protocols (static route to R5's LAN + RIP redistribution).

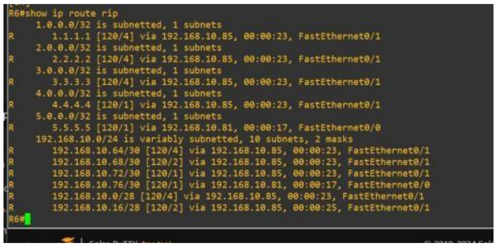


Figure 8.20: R6 – show ip route rip (RIP routes across the full topology).

10. 9. Troubleshooting Report (Real Issues Encountered During the Build)

The following issues were encountered and resolved while building and verifying this topology in GNS3. Each is documented with its symptom, diagnostic method, and fix, as evidence of systematic troubleshooting.

9.1 R3-R4 link: no connectivity despite correct IP configuration

Symptom: R3 could not ping R4's directly-connected interface (0% success), even though both interfaces showed "up/up" with correct IP addresses on the same /30 subnet.

Diagnosis: "show cdp neighbors" on both routers revealed the physical cable was connected to the wrong port — R3's Fa1/0 (intended for the LAN3 switch) was wired to R4's Fa0/0, while R3's actual WAN interface (Fa0/1) was unconnected.

Fix: Deleted the mis-wired link in the GNS3 canvas and reconnected R3's Fa0/1 to R4's Fa0/0, then reconnected Switch2 to R3's Fa1/0. Flapped both interfaces (shutdown/no shutdown) and confirmed 100% ping success.

9.2 RIP routes not propagating between R3 and R4

Symptom: After fixing the cabling, "show ip route rip" on R3 remained empty even though the R3-R4 link was reachable.

Diagnosis: "show running-config" confirmed R3 was still running RIP v1 (Phase A) while R4 had already been upgraded to RIP v2 (Phase B) — version mismatch prevented the routers from accepting each other's updates.

Fix: Applied "version 2" and "no auto-summary" to R3's RIP process to match R4 and R6. Routes began appearing within one update interval.

9.3 R5-R6 link: one-directional (simplex) connectivity

Symptom: R4 could ping R6 successfully, but R5 could not ping R6's neighboring interface even though both interfaces were "up/up", had correct IP/mask, and CDP showed a valid adjacency. "show interfaces" on R5 showed 0 packets ever received on that interface, while R6's side showed packets received.

Diagnosis: This indicated a corrupted/simplex virtual link in the GNS3 canvas rather than a configuration issue, since all IP-level settings matched.

Fix: Deleted the link between R5 and R6 in the GNS3 canvas and recreated it fresh, then flapped both interfaces. Connectivity was restored (confirmed with an 80% then 100% ping success rate).

9.4 End devices (VPCS) losing IP configuration

Symptom: Several PCs (PC1, PC3, PC7, PC8) intermittently returned "host not reachable" while the routers' routing tables were fully converged and correct.

Diagnosis: "show ip" on the affected VPCS nodes showed IP/MASK and GATEWAY reset to 0.0.0.0 — the static IP configuration had not been saved and was lost on a VPCS restart.

Fix: Re-applied the static IP/gateway on each affected PC (e.g. "ip 192.168.10.2/28 192.168.10.1") and used "save" to persist the configuration across restarts.

11. 10. Challenges Faced

The main challenges in this project were not with the routing protocol logic itself, but with correctly wiring a six-router, four-switch, eight-PC topology in GNS3 — a single mis-connected cable between two ports with matching subnet numbering was difficult to detect from routing table output alone, and required systematic use of "show cdp neighbors" to confirm the physical topology matched the intended design. RIP version mismatches (v1 vs v2) also silently blocked route exchange without any error message, requiring explicit verification of "show runningconfig" on both ends of each RIP-enabled link.

12. 11. Conclusion

This project successfully implemented and verified a multi-protocol enterprise network combining OSPF Multi-Area routing, RIP v1/v2, and static routing, integrated through route redistribution across two redistribution points (R3 for OSPF<->RIP, and R6 for Static->RIP). End-to-end connectivity was confirmed from all four LAN segments to each other. The troubleshooting process — particularly resolving cabling mismatches and a RIP version mismatch — reinforced the importance of verifying both the physical topology (via CDP) and protocol-level configuration when diagnosing routing failures in an enterprise network.